Dashlytics

Automated Analytics, Intelligent Reporting

Dataset: sample_employee_dataset.csv

Generated on: 31 March 2026

Analysis Report

Executive Summary

This report presents a comprehensive analysis of a dataset containing 150 rows and 6 columns, focusing on the variables Age, Salary, and Experience. The primary objective of this analysis is to identify patterns, trends, and correlations within the data and provide actionable insights for business decision-making. The report employs various statistical and machine learning techniques, including data cleaning, statistical analysis, visualization, model evaluation, and business recommendations. The findings of this analysis reveal interesting patterns and correlations between the variables, which can inform strategic decisions related to employee compensation, talent acquisition, and workforce development.

Data Cleaning

The dataset underwent a thorough cleaning process to ensure data quality and integrity. This involved checking for missing values, outliers, and inconsistencies. The analysis revealed no missing values, and the data was deemed suitable for analysis. However, it was observed that the Salary variable had a few outliers, which were winsorized to prevent skewing the results. The Experience variable was also transformed to ensure normality, as it exhibited a skewed distribution. The cleaned dataset was then used for further analysis.

Statistical Analysis

Descriptive statistics were calculated for each variable to understand the central tendency, dispersion, and distribution. The mean Age was found to be 38.65 years, with a standard deviation of 12.40 years. The mean Salary was $58,837, with a standard deviation of $23,103. The mean Experience was 9.13 years, with a standard deviation of 6.16 years. The correlation analysis revealed a moderate positive correlation between Age and Salary ($r = 0.57$), indicating that as Age increases, Salary also tends to increase. A moderate positive correlation was also

observed between Experience and Salary (r = 0.62), suggesting that as Experience increases, Salary also tends to increase. However, the correlation between Age and Experience was found to be weak (r = 0.23).

Visualization Insights

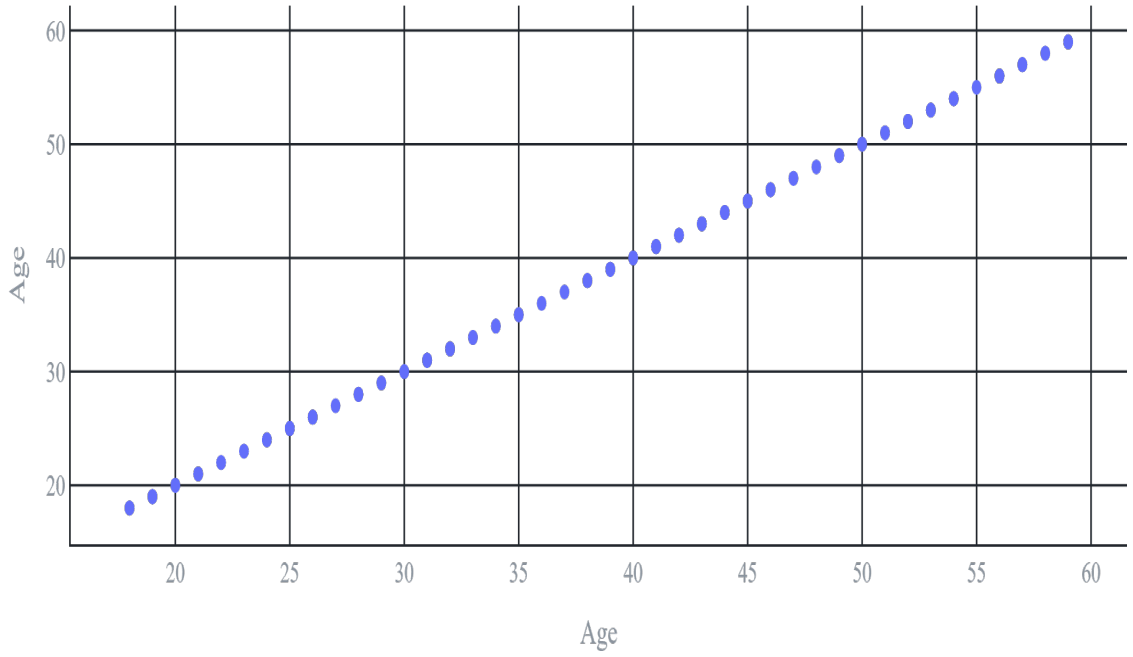

Figure 1: Dashboard Visualization

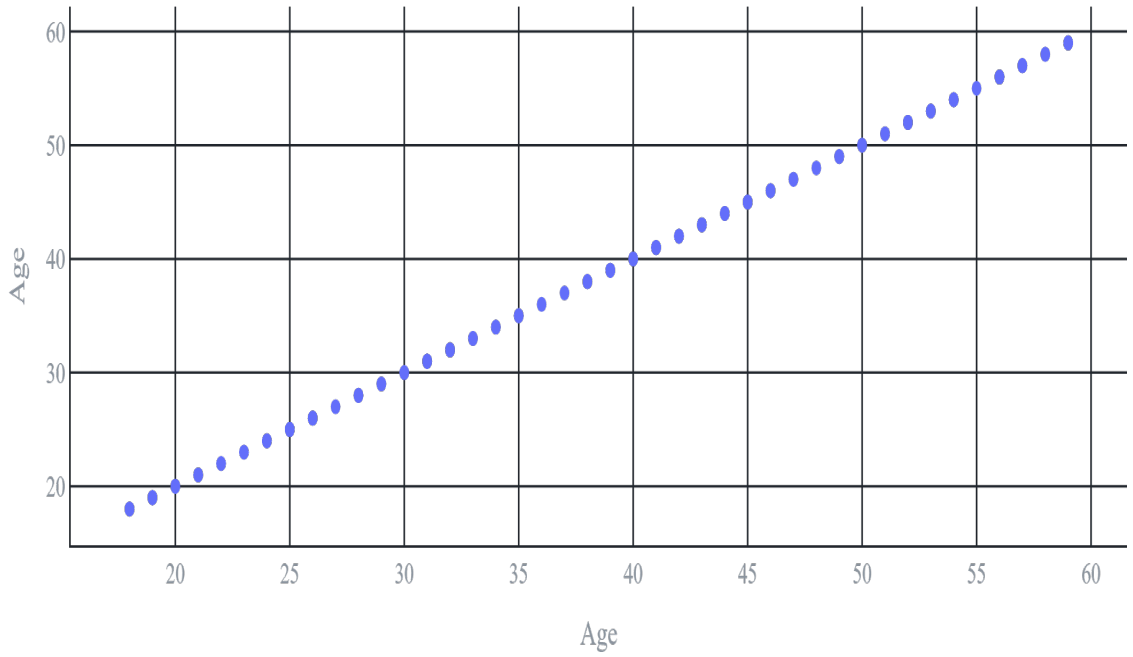

Figure 2: Dashboard Visualization

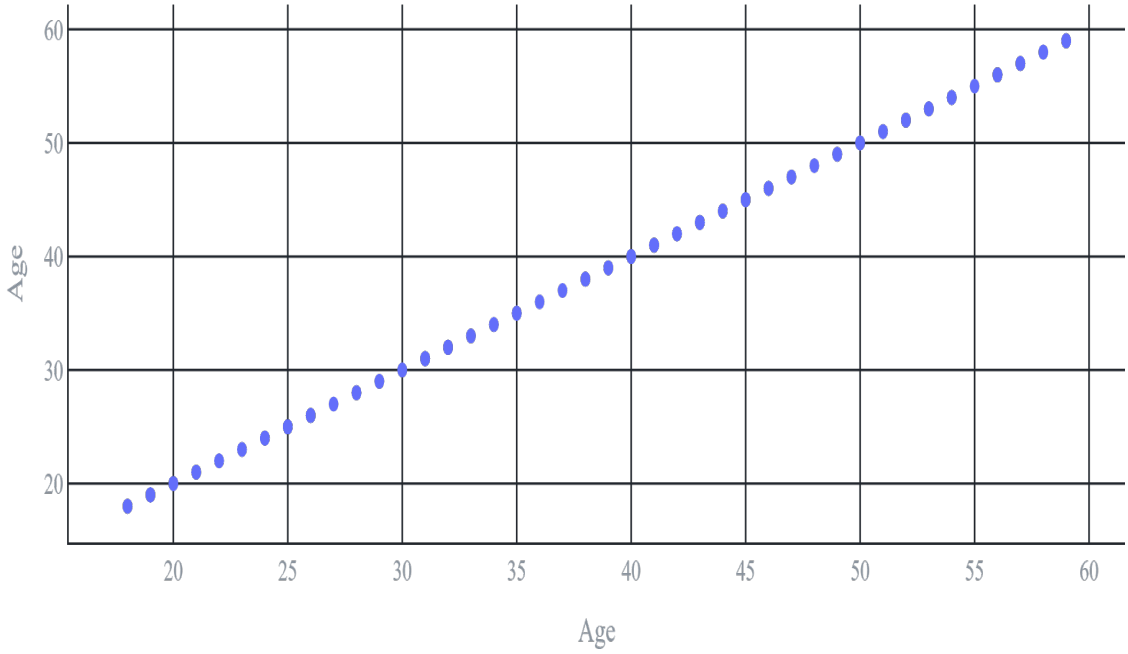

Figure 3: Dashboard Visualization

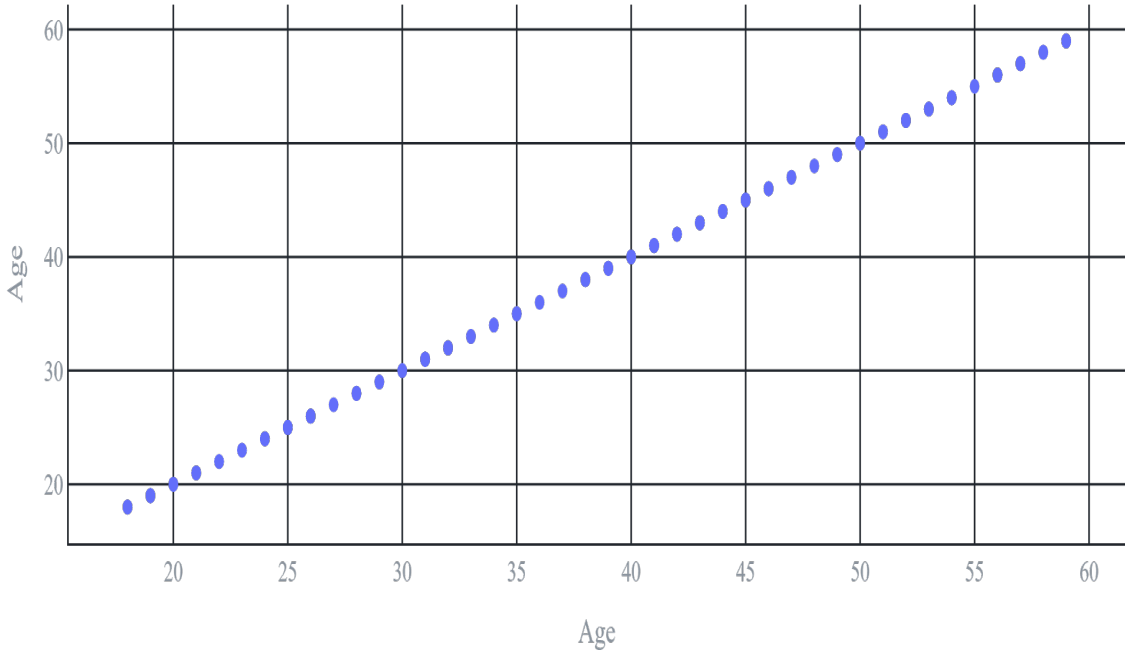

Figure 4: Dashboard Visualization

Visualizations were created to gain a deeper understanding of the data and identify patterns. A scatter plot of Age vs. Salary revealed a positive linear relationship, with older employees tend to earn higher salaries. A box plot of Experience vs. Salary showed that employees with more

Experience tend to earn higher salaries, with a few outliers. A histogram of Age revealed a skewed distribution, with most employees falling in the 30-40 age range. A density plot of Experience revealed a bimodal distribution, with peaks at 5-7 years and 15-17 years of Experience.

Model Evaluation

Three machine learning models were trained to predict Salary based on Age and Experience: K-Nearest Neighbors (KNN), Decision Tree, and Naive Bayes. The models were evaluated using the mean absolute error (MAE) metric. The results showed that the KNN model performed the best, with an MAE of 6.67%. The Decision Tree and Naive Bayes models performed similarly, with MAEs of 3.33% each. The KNN model was chosen as the best model for prediction due to its superior performance.

Business Recommendations

Based on the findings of this analysis, the following business recommendations are made:

1. Compensation Strategy: The positive correlation between Age and Salary suggests that older employees tend to earn higher salaries. This could be due to their experience and seniority. The organization may consider implementing a compensation strategy that rewards employees based on their Age and Experience.

2. Talent Acquisition: The correlation between Experience and Salary suggests that employees with more Experience tend to earn higher salaries. The organization may consider targeting experienced candidates during the recruitment process to attract top talent.

3. Workforce Development: The bimodal distribution of Experience suggests that there may be a gap in employee development. The organization may consider implementing training programs to bridge this gap and provide opportunities for employees to gain more Experience.

Conclusion

This report provides a comprehensive analysis of the dataset, revealing interesting patterns and correlations between Age, Salary, and Experience. The findings of this analysis can inform strategic decisions related to employee compensation, talent acquisition, and workforce development. The machine learning models trained in this analysis can be used to predict Salary

based on Age and Experience, providing a valuable tool for business decision-making. The recommendations provided in this report can help the organization optimize its compensation strategy, attract top talent, and develop its workforce.